

Predictive & Corrective Maintenance Rules — Production Line A

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1. Predictive Maintenance Rules

Rule ID	Sensor	Station	Trigger	Action	Priority
MAINT-01	VIB	ST04_PACKAGING	VIB DRIFT >0.05 mm/s over 60 min	Inspect and lubricate bearings	HIGH
MAINT-02	CUR	ST02_SEALING	CUR DRIFT >1.0 A for 30+ min	Inspect sealing jaw	HIGH
MAINT-03	TEN	ST03_LABELLING	TEN STUCK >10 samples	Replace sensor	MEDIUM
MAINT-04	PRS	ST01_FILLING	PRS <3.3 bar for >15 min	Inspect filter and pump	MEDIUM
MAINT-05	TMP	SRV01_SERVERROOM	TMP >25°C for >5 min	Activate secondary cooling	CRITICAL
MAINT-06	TMP	WRH01_WAREHOUSE	TMP DRIFT >2°C over 2 h	Inspect compressor; notify QC	CRITICAL
MAINT-07	TMP	CHM01_CHEMICALSTORAGE	TMP >20°C for >10 min	Emergency ventilation; notify Safety Officer	CRITICAL
MAINT-08	HUM	RND01_RDLAB	HUM DRIFT >10%RH over 60 min	Activate dehumidifier; check HVAC drain	HIGH

2. Correlated Fault Resolution

Pattern	Primary Fault	Secondary Effect	Resolution
ST02_SEALING-CUR↑ → ST04_PACKAGING-SPEED↓	Sealing pressure loss	Conveyor speed drop via electrical stop	1. Verify ST02 CUR trend. 2. Inspect sealing jaw. 3. Reset ST04 speed.
SRV01-TMP↑ → SCADA latency↑	Server Room overheating	Increased scan time → apparent STUCK	1. Reduce SRV01 TMP <25°C. 2. Verify PLC heartbeats. 3. Clear STUCK.
WRH01-TMP↑ → QC HOLD	Cold warehouse drift	Product quality risk	1. Restore WRH01-TMP <8°C. 2. Notify QC. 3. Place batch on HOLD.